Gene Set Enrichment Report

Generated by direct API calls to g:Profiler, Enrichr, and STRING. Species: *Mus musculus* (taxid 10090 / mmusculus). Raw tool output only — no interpretation applied.

Input gene list (n=25):
`Slc30a9, Cd68, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Lrrfip2, Gjb3, Atp1b3, Mms19, Mical2, Zbtb34, Ddit3, Slc26a11, Hdac2, Mif, Gli3, Dst, Dtnbp1, Mafg, Tnfsf18`

0. Simple summary (top 5 terms per source)

A flattened, easy-scan version of the detailed tables below. Same underlying numbers, fewer columns.

Table form

Source	Term
g:Profiler (GO:MF)	protein binding

Source	Term
g:Profiler (GO:BP)	negative regulation of alpha-beta T cell differentiation
g:Profiler (GO:BP)	negative regulation of alpha-beta T cell activation
g:Profiler (GO:BP)	negative regulation of T cell differentiation
g:Profiler (GO:BP)	positive regulation of macromolecule metabolic process
g:Profiler (GO:BP)	negative regulation of lymphocyte differentiation
g:Profiler (GO:CC)	cytoplasm

Source	Term
Enrichr — GO:BP	negative regulation of cellular macromolecule biosynthetic process (GO:2000
Enrichr — GO:BP	regulation of heart contraction (GO:0008016)
Enrichr — GO:BP	embryonic digit morphogenesis (GO:0042733)
Enrichr — GO:BP	positive regulation of nucleic acid-templated transcription (GO:1903508)
Enrichr — GO:BP	positive regulation of transcription by RNA polymerase II (GO:0045944)
Enrichr — GO:MF	LRR domain binding (GO:0030275)
Enrichr — GO:MF	nuclear receptor coactivator activity (GO:0030374)
Enrichr — GO:MF	DNA-binding transcription factor binding (GO:0140297)
Enrichr — GO:MF	histone deacetylase binding (GO:0042826)
Enrichr — GO:MF	cytokine receptor binding (GO:0005126)
Enrichr — GO:CC	intracellular membrane-bounded organelle (GO:0043231)

Source	Term
Enrichr — GO:CC	nucleus (GO:0005634)
Enrichr — GO:CC	P-body (GO:0000932)
Enrichr — GO:CC	microtubule cytoskeleton (GO:0015630)
Enrichr — GO:CC	hemidesmosome (GO:0030056)
Enrichr — KEGG (mouse)	Kaposi sarcoma-associated herpesvirus infection
Enrichr — KEGG (mouse)	Thyroid hormone signaling pathway
Enrichr — KEGG (mouse)	Apoptosis
Enrichr — KEGG (mouse)	Non-alcoholic fatty liver disease (NAFLD)
Enrichr — KEGG (mouse)	Oxytocin signaling pathway
Enrichr — WikiPathways (mouse)	Calcium Regulation in the Cardiac Cell WP553

Source	Term
Enrichr — WikiPathways (mouse)	IL-5 Signaling Pathway WP151
Enrichr — WikiPathways (mouse)	Estrogen signaling WP1244
Enrichr — WikiPathways (mouse)	Delta-Notch Signaling Pathway WP265
Enrichr — WikiPathways (mouse)	Histone modifications WP300
Enrichr — MGI Phenotype	abnormal olfactory nerve morphology MP:0005236
Enrichr — MGI Phenotype	embryonic lethality during organogenesis, incomplete penetrance MP:001110
Enrichr — MGI Phenotype	abnormal neural tube morphology MP:0002151
Enrichr — MGI Phenotype	eyelids open at birth MP:0001302
Enrichr — MGI Phenotype	abnormal heart morphology MP:0000266

List form

g:Profiler — GO:MF

- **protein binding** — p_adj=9.80e-06 — genes: Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Lrrfip2, Gjb3, Atp1b3, Mms19, Mical2, Zbtb34, Hdac2, Mif, Gli3, Dst, Dtnbp1, Mafg, Tnfsf18

g:Profiler — GO:BP

- **negative regulation of alpha-beta T cell differentiation** — p_adj=2.00e-05 — genes: Rc3h1, Lgals1, Gli3, Tnfsf18
- **negative regulation of alpha-beta T cell activation** — p_adj=1.18e-04 — genes: Rc3h1, Lgals1, Gli3, Tnfsf18

- **negative regulation of T cell differentiation** — p_adj=1.53e-04 — genes: Rc3h1, Lgals1, Gli3, Tnfsf18
- **positive regulation of macromolecule metabolic process** — p_adj=2.12e-04 — genes: Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Jun, Mms19, Mical2, Hdac2, Mif, Gli3, Dtnbp1, Mafg
- **negative regulation of lymphocyte differentiation** — p_adj=2.46e-04 — genes: Rc3h1, Lgals1, Gli3, Tnfsf18

g:Profiler — GO:CC

- **cytoplasm** — p_adj=1.75e-03 — genes: Slc30a9, Cd68, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Gjb3, Atp1b3, Mms19, Mical2, Slc26a11, Hdac2, Mif, Gli3, Dst, Dtnbp1

Enrichr — GO:BP

- **negative regulation of cellular macromolecule biosynthetic process (GO:2000113)** — p_adj=2.43e-02 — genes: JUN, GSK3A, HDAC2, DDIT3, CNOT9, GLI3
- **regulation of heart contraction (GO:0008016)** — p_adj=3.96e-02 — genes: GSK3A, ITPR2, ATP1B3
- **embryonic digit morphogenesis (GO:0042733)** — p_adj=3.96e-02 — genes: HDAC2, GLI3
- **positive regulation of nucleic acid-templated transcription (GO:1903508)** — p_adj=4.87e-02 — genes: JUN, HDAC2, DDIT3, MMS19, GLI3
- **positive regulation of transcription by RNA polymerase II (GO:0045944)** — p_adj=6.89e-02 — genes: SLC30A9, JUN, HDAC2, DDIT3, MICAL2, GLI3

Enrichr — GO:MF

- **LRR domain binding (GO:0030275)** — p_adj=1.18e-02 — genes: DDIT3, LRRFIP2
- **nuclear receptor coactivator activity (GO:0030374)** — p_adj=6.25e-02 — genes: SLC30A9, CNOT9
- **DNA-binding transcription factor binding (GO:0140297)** — p_adj=6.25e-02 — genes: JUN, HDAC2, DDIT3
- **histone deacetylase binding (GO:0042826)** — p_adj=7.42e-02 — genes: HDAC2, GLI3
- **cytokine receptor binding (GO:0005126)** — p_adj=7.42e-02 — genes: TNFSF18, MIF

Enrichr — GO:CC

- **intracellular membrane-bounded organelle (GO:0043231)** — p_adj=8.69e-02 — genes: SLC30A9, JUN, GSK3A, HDAC2, DST, DTNBP1, MICAL2, GLI3, RCAN1, GJB3, MAFG, DDIT3, MMS19, SLC26A11
- **nucleus (GO:0005634)** — p_adj=8.69e-02 — genes: RCAN1, SLC30A9, JUN, GSK3A, HDAC2, DST, MAFG, DDIT3, DTNBP1, MICAL2, MMS19, GLI3

- **P-body (GO:0000932)** — p_adj=8.69e-02 — genes: RC3H1, CNOT9
- **microtubule cytoskeleton (GO:0015630)** — p_adj=8.69e-02 — genes: DST, DTNBP1, MMS19
- **hemidesmosome (GO:0030056)** — p_adj=8.69e-02 — genes: DST

Enrichr — KEGG (mouse)

- **Kaposi sarcoma-associated herpesvirus infection** — p_adj=1.51e-02 — genes: RCAN1, JUN, ITPR2, GABARAP
- **Thyroid hormone signaling pathway** — p_adj=1.96e-02 — genes: RCAN1, HDAC2, ATP1B3
- **Apoptosis** — p_adj=1.96e-02 — genes: JUN, DDIT3, ITPR2
- **Non-alcoholic fatty liver disease (NAFLD)** — p_adj=1.96e-02 — genes: JUN, GSK3A, DDIT3
- **Oxytocin signaling pathway** — p_adj=1.96e-02 — genes: RCAN1, JUN, ITPR2

Enrichr — WikiPathways (mouse)

- **Calcium Regulation in the Cardiac Cell WP553** — p_adj=3.18e-02 — genes: GJB3, ITPR2, ATP1B3
- **IL-5 Signaling Pathway WP151** — p_adj=4.80e-02 — genes: JUN, GSK3A
- **Estrogen signaling WP1244** — p_adj=4.80e-02 — genes: JUN, HDAC2
- **Delta-Notch Signaling Pathway WP265** — p_adj=4.80e-02 — genes: JUN, HDAC2
- **Histone modifications WP300** — p_adj=4.99e-02 — genes: HDAC2

Enrichr — MGI Phenotype

- **abnormal olfactory nerve morphology MP:0005236** — p_adj=3.88e-02 — genes: LGALS1, GLI3
- **embryonic lethality during organogenesis, incomplete penetrance MP:0011108** — p_adj=3.88e-02 — genes: JUN, HDAC2, GJB3, GLI3
- **abnormal neural tube morphology MP:0002151** — p_adj=7.91e-02 — genes: JUN, ATP1B3, GLI3
- **eyelids open at birth MP:0001302** — p_adj=7.91e-02 — genes: JUN, GLI3
- **abnormal heart morphology MP:0000266** — p_adj=7.91e-02 — genes: RCAN1, JUN, HDAC2, GJB3

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1. g:Profiler (g:GOSt)

Sources queried: GO:BP, GO:MF, GO:CC, KEGG, Reactome, WikiPathways. Correction method: g:SCS (default). "Adjusted p-value" below is g:Profiler's native corrected p_value output. **Note:** `Ddit3` maps ambiguously to two

Ensembl mouse gene IDs (ENSMUSG00000025408 and ENSMUSG00000116429) and was excluded from this run's statistics as a result — effective query size 23–24 depending on term. All other 24 genes mapped uniquely.

Source	Term ID	Term name	Adj. p-value	T
GO:MF	GO:0005515	protein binding	9.804e-06	9
GO:BP	GO:0046639	negative regulation of alpha-beta T cell differentiation	2.004e-05	3
GO:BP	GO:0046636	negative regulation of alpha-beta T cell activation	1.182e-04	4
GO:BP	GO:0045581	negative regulation of T cell differentiation	1.532e-04	4
GO:BP	GO:0010604	positive regulation of macromolecule metabolic process	2.121e-04	2
GO:BP	GO:0045620	negative regulation of lymphocyte differentiation	2.456e-04	5
GO:BP	GO:2000320	negative regulation of T-helper 17 cell differentiation	5.146e-04	1
GO:BP	GO:0009893	positive regulation of metabolic process	5.403e-04	3
GO:BP	GO:2000317	negative regulation of T-helper 17 type immune response	6.245e-04	1
GO:BP	GO:0046637	regulation of alpha-beta T cell differentiation	7.333e-04	7

Source	Term ID	Term name	Adj. p-value	T
GO:BP	GO:0051254	positive regulation of RNA metabolic process	1.060e-03	1
GO:BP	GO:0048522	positive regulation of cellular process	1.301e-03	5
GO:CC	GO:0005737	cytoplasm	1.746e-03	1
GO:BP	GO:0045623	negative regulation of T-helper cell differentiation	1.851e-03	2
GO:BP	GO:0043371	negative regulation of CD4-positive, alpha-beta T cell differentiation	2.375e-03	2
GO:BP	GO:2000319	regulation of T-helper 17 cell differentiation	2.670e-03	2
GO:BP	GO:0010557	positive regulation of macromolecule biosynthetic process	3.047e-03	2
GO:BP	GO:1902106	negative regulation of leukocyte differentiation	3.214e-03	1
GO:BP	GO:0048518	positive regulation of biological process	3.456e-03	5

Source	Term ID	Term name	Adj. p-value	T
GO:BP	GO:1903707	negative regulation of hemopoiesis	4.021e-03	1

2. Enrichr

Gene list submitted as-is (mouse gene symbols); Enrichr internally uppercases symbols for matching against its libraries. Top 8 terms per library shown, ranked by adjusted p-value (Benjamini-Hochberg).

GO Biological Process (2021)

Term	p-value	Adjusted p-valu
negative regulation of cellular macromolecule biosynthetic process (GO:2000113)	4.629e-05	2.426e-02
regulation of heart contraction (GO:0008016)	2.079e-04	3.960e-02
embryonic digit morphogenesis (GO:0042733)	2.267e-04	3.960e-02
positive regulation of nucleic acid-templated transcription (GO:1903508)	3.714e-04	4.866e-02
positive regulation of transcription by RNA polymerase II (GO:0045944)	7.257e-04	6.892e-02
axonal transport (GO:0098930)	9.207e-04	6.892e-02
embryonic limb morphogenesis (GO:0030326)	9.207e-04	6.892e-02
negative regulation of canonical Wnt signaling pathway (GO:0090090)	1.110e-03	7.163e-02

GO Molecular Function (2021)

Term	p-value	Adjusted p-value	Odds ratio
LRR domain binding (GO:0030275)	1.353e-04	1.177e-02	144.66
nuclear receptor coactivator activity (GO:0030374)	1.988e-03	6.246e-02	33.97

Term	p-value	Adjusted p-value	Odds ratio
DNA-binding transcription factor binding (GO:0140297)	2.154e-03	6.246e-02	13.15
histone deacetylase binding (GO:0042826)	5.986e-03	7.415e-02	19.00
cytokine receptor binding (GO:0005126)	7.570e-03	7.415e-02	16.78
sequence-specific DNA binding (GO:0043565)	1.084e-02	7.415e-02	5.22
sequence-specific double-stranded DNA binding (GO:1990837)	1.111e-02	7.415e-02	5.18
cAMP response element binding protein binding (GO:0008140)	1.120e-02	7.415e-02	103.99

GO Cellular Component (2021)

Term	p-value	Adjusted p-value	Odds ratio	Comb
intracellular membrane-bounded organelle (GO:0043231)	1.358e-03	8.692e-02	3.64	24.01
nucleus (GO:0005634)	4.242e-03	8.692e-02	3.20	17.48
P-body (GO:0000932)	4.465e-03	8.692e-02	22.18	120.0
microtubule cytoskeleton (GO:0015630)	7.888e-03	8.692e-02	8.17	39.55
hemidesmosome (GO:0030056)	8.718e-03	8.692e-02	138.67	657.6
ESC/E(Z) complex (GO:0035098)	9.958e-03	8.692e-02	118.86	547.8
platelet dense tubular network membrane (GO:0031095)	1.120e-02	8.692e-02	103.99	467.1
apical dendrite (GO:0097440)	1.243e-02	8.692e-02	92.44	405.5

KEGG (2019, Mouse)

Term	p-value	Adjusted p-value	Odds ratio	Combined sc
Kaposi sarcoma-associated herpesvirus infection	1.401e-04	1.513e-02	17.76	157.56
Thyroid hormone signaling pathway	3.884e-04	1.964e-02	24.18	189.93
Apoptosis	7.041e-04	1.964e-02	19.60	142.28
Non-alcoholic fatty liver disease (NAFLD)	8.589e-04	1.964e-02	18.27	128.97
Oxytocin signaling pathway	9.092e-04	1.964e-02	17.90	125.37
NOD-like receptor signaling pathway	2.067e-03	3.461e-02	13.35	82.52
cAMP signaling pathway	2.243e-03	3.461e-02	12.96	79.05
Mitophagy	2.796e-03	3.774e-02	28.39	166.91

WikiPathways (2019, Mouse)

Term	p-value	Adjusted p-value	Odds
Calcium Regulation in the Cardiac Cell WP553	7.946e-04	3.179e-02	18.78
IL-5 Signaling Pathway WP151	3.343e-03	4.798e-02	25.84
Estrogen signaling WP1244	4.144e-03	4.798e-02	23.07
Delta-Notch Signaling Pathway WP265	4.798e-03	4.798e-02	21.36
Histone modifications WP300	6.235e-03	4.988e-02	208.0
Adipogenesis genes WP447	1.208e-02	6.216e-02	13.07
Macrophage markers WP2271	1.243e-02	6.216e-02	92.44
Ethanol metabolism resulting in production of ROS by CYP2E1 WP4265	1.243e-02	6.216e-02	92.44

MGI Mammalian Phenotype Level 4 (2021)

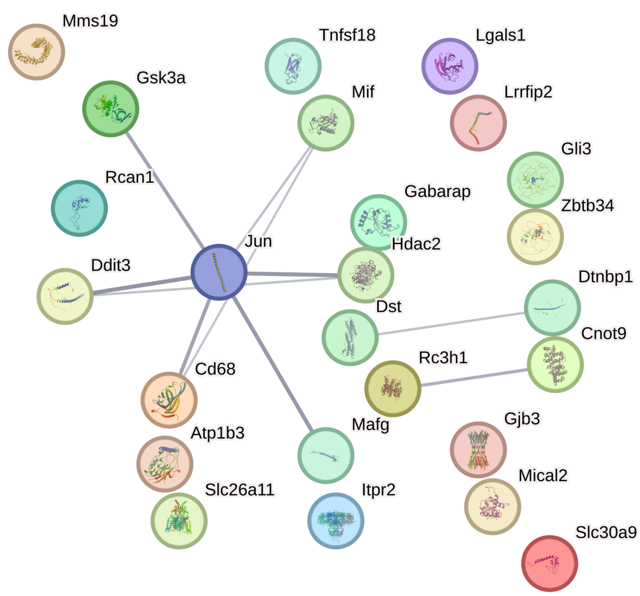
Term	p-value	Adjusted p-value
abnormal olfactory nerve morphology MP:0005236	1.781e-04	3.882e-02

Term	p-value	Adjusted p-value
embryonic lethality during organogenesis, incomplete penetrance MP:0011108	2.065e-04	3.882e-02
abnormal neural tube morphology MP:0002151	1.542e-03	7.907e-02
eyelids open at birth MP:0001302	1.914e-03	7.907e-02
abnormal heart morphology MP:0000266	2.883e-03	7.907e-02
abnormal myocardium layer morphology MP:0005329	4.465e-03	7.907e-02
decreased fibroblast proliferation MP:0011704	5.258e-03	7.907e-02
oxidative stress MP:0003674	6.625e-03	7.907e-02

3. STRING

Network built at STRING's default confidence threshold (medium, 0.4) using all active interaction sources (neighborhood, fusion, cooccurrence, coexpression, experiments, database, textmining), species restricted to *Mus musculus* (taxid 10090). All 25 input symbols resolved to a unique STRING identifier (no ambiguous or failed mappings).

3a. Network image



3b. Interaction edges (returned by STRING)

Score columns are STRING's per-channel confidence scores (0–1): nscore=neighborhood, fscore=fusion, pscore=phylogenetic cooccurrence, ascore=coexpression, escore=experimental, dscore=database, tscore=textmining. "score" is the combined score.

Gene A	Gene B	Combined score	nscore	fscore	pscore	ascore	escore	dscore	tscore
Cd68	Mif	0.405	0	0	0	0.066	0	0	0.39
Cd68	Jun	0.642	0	0	0	0	0	0	0.642
Hdac2	Ddit3	0.424	0	0	0	0.047	0.132	0	0.36
Hdac2	Jun	0.707	0	0	0	0	0.096	0	0.69
Ddit3	Jun	0.722	0	0	0	0.077	0.123	0	0.685
Mif	Jun	0.43	0	0	0	0	0.065	0	0.415
Mafg	Jun	0.699	0	0	0	0.057	0	0	0.694
Gsk3a	Jun	0.627	0	0	0	0	0.247	0	0.525
Dtnbp1	Dst	0.424	0	0	0	0	0	0	0.425
Cnot9	Rc3h1	0.553	0	0	0	0	0.423	0	0.258

3c. Per-gene connectivity (degree in returned network)

Gene	Degree (# edges)
Slc30a9	0
Cd68	2
Rc3h1	1
Cnot9	1
Gsk3a	1
Gabarap	0
Rcan1	0
Itpr2	0
Jun	6
Lgals1	0
Lrrfip2	0
Gjb3	0
Atp1b3	0

Gene	Degree (# edges)
Mms19	0
Mical2	0
Zbtb34	0
Ddit3	2
Slc26a11	0
Hdac2	2
Mif	2
Gli3	0
Dst	1
Dtnbp1	1
Mafg	1
Tnfsf18	0

3d. STRING functional enrichment (top 15 by FDR, across all categories returned)

Category	Term ID	Description	# genes
Process	GO:0010604	Positive regulation of macromolecule metabolic process	16
Process	GO:0031325	Positive regulation of cellular metabolic process	15
Process	GO:0051173	Positive regulation of nitrogen compound metabolic process	15
COMPARTMENTS	GOCC:0005737	Cytoplasm	20
Process	GO:0031323	Regulation of cellular metabolic process	18
Process	GO:0046639	Negative regulation of alpha-beta T cell differentiation	3
Process	GO:0048583	Regulation of response to stimulus	15
Process	GO:0051171	Regulation of nitrogen compound metabolic process	18
Process	GO:0051254	Positive regulation of RNA metabolic process	11
Process	GO:0080090	Regulation of primary metabolic process	18
Keyword	KW-0678	Repressor	6
Process	GO:0010468	Regulation of gene expression	16
Process	GO:0045893	Positive regulation of transcription, DNA-templated	10

Category	Term ID	Description	# genes
Process	GO:0048518	Positive regulation of biological process	18
Process	GO:0048522	Positive regulation of cellular process	17

Data provenance

g:Profiler: `POST https://biit.cs.ut.ee/gprofiler/api/gost/profile/`
Enrichr: `POST https://maayanlab.cloud/Enrichr/addList` + `GET /Enrichr/enrich`
STRING: `GET https://string-db.org/api/tsv/get_string_ids`, `/network`, `/enrichment`, `/highres_image/network`
All calls made live against each service; no cached or precomputed data used. Report generated 2026-08-11 15:28.